

Task 3: Concept Space & PCA Steering (Encoder-Decoder)

1. Objective

This task aims to extract interpretable latent directions from the model's hidden states using Principal Component Analysis (PCA). By shifting the representation along the "Diversity PC", we attempt to control the generation variance without retraining.

Key Metric: The Encoder-Decoder variant captures an exceptionally high ****90.9% variance**** in the first 50 components, suggesting a very structured (though potentially rigid) latent space.

90.9%

Explained Variance

$|r|=0.56$

Diversity Correlation

0.222

Semantic Stability

2. Technical Architecture

Hidden representations are extracted from the encoder-decoder bottleneck across a set of 120 samples. Steering is applied via the formula: $h_{\text{steered}} = h + \alpha \cdot v_{\text{diversity}}$.

👉 **Implementation:** [analysis/concept_vectors.py](#)

3. Diversity Spectrum (Output Samples)

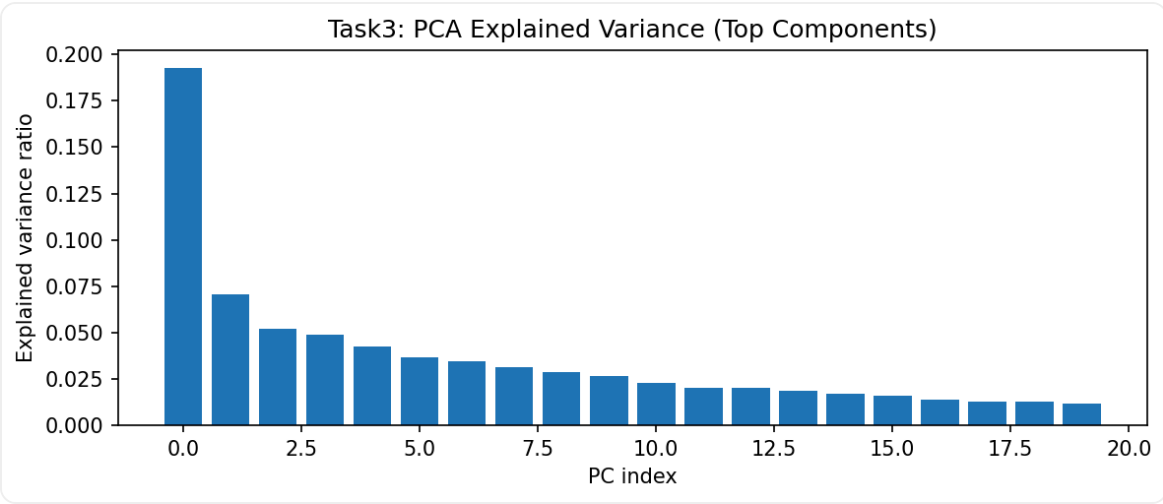
Analysis of the diversity spectrum shows significant token repetition ("निर्यात" and "नेमि") at extreme alpha values, indicating a collapse in syntax under strong steering.

```
α = -2.0: निर्यात निर्यात निर्यात (Repetitive)
α = +0.0: धर्मो रक्षति रक्षितः (Baseline)
α = +2.0: नेमि नेमि नेमि नेमि ... (Repetitive)
```

4. Visual Diagnostics

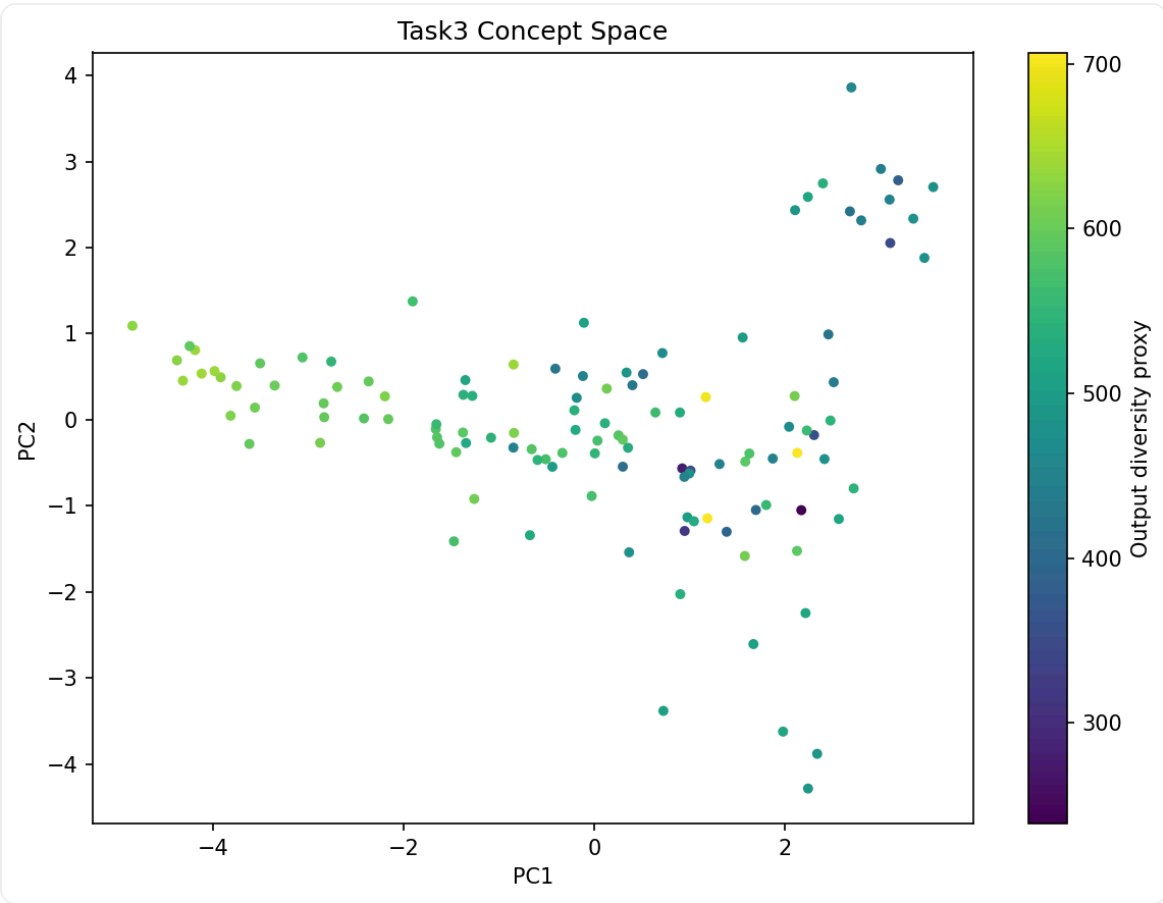
4.1 PCA Explained Variance

The "elbow" in the variance curve is sharp, with the first 5 components capturing over 60% of information.



4.2 Concept Space Projection

Clusters in the latent space appear focused but show significant overlap between different semantic regimes.



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